22/9/22 Exp. 6 Hamming code

Aim

Write a program to implement error correction and detection using hamming code

Code :

```
def - calc - redundant bits (m):
        for r in range (m+1):
            if (2**r) >= (m+r+1):
                return r

def main ():
    data = input ("Enter data bits: ").strip()
    m = len(data)
    r = calc - redundant - bits (m)
    n = m+r
    print (f" number of r needed : {r}")
    print (f" Total frame length : {n}")
    code = ['0' for i in (m+1)]
    j = 0
    for i in range (1, n+1):
        if (i & (i-1)) != 0:
            code [i] = data [-j-1]
            j += 1

    for i in range (r):
        parity-pos = 2**i
        val = 0
        for pos in range (1, n+1):
            if (pos & parity-pos) != 0 and
```

```
if code[pos-1]=='1':
    val^=1
code[parity_pos-1]=str(val)
transmitted_str="".join(code[1:][::-1])
print(f"Data to be transmitted: {transmitted_str}")
change=input("In Transmission bit change").strip().
if changed=="yes":
    received_str=input(f"enter corrupted data").strip()
    if(len(received_str)!=n:
        print(f"error: received data is exactly {n} bits
                long!").
        return
    received_code=['0']+list(received_str[::-1])
else:
    received_code=list(code)
error_position=0
for i in range(r):
    parity_pos=2**i
    val=0
    for pos in range(1,n+1):
        if(pos & parity_pos)!=0:
            if(received_code[pos])=='1':
                val^=1
    if val!=0:
        error_position+=parity_pos
```

```
if error_position == 0:
    print("\n Verification result: no error occured")
else
    print("\n Verification result = error detected !")
    print(f" wrong bit at position: {error_position}")
    received_code[error_position] = '0' if received_code
        [error_position] == '1' else '1'
    corrected = ''.join(received_code[1:][::-1])
    print(f" corrected data stream: {corrected_str}")
if __name__ == "__main__":
    main()
```

output:

① Enter data bits: 10110

number of redundant bits (r) needed: 4

total frame length (m+r): 10

data to be transmitted: 1011100011

Transmitted bit change? (yes/no): yes

Enter the corrupted data stream (10 bits):
1011101011

verification result: Error detected!

wrong bit is at position: 7

corrected data stream: 1011100011

(2)

Enter data bits : 1011

number of redundant bits (r) needed : 3

total data frame length (m+r) : 7

data to be transmitted : (01610)

Transmitted bit change ? (Yes/No) : No

Verification Result : no error detected in transmission

Result :

Thus the hamming code was implemented successfully for error detection and correction